

A Review of Face Anti-spoofing Methods for Face Recognition Systems

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Abstract— The applicability, popularity, and usage of face recognition systems are undoubtedly increasing due to their advantages of being very convenient, contactless, and non-intrusive when compared with other biometric systems such as voice recognition and fingerprint. Unfortunately, it is the most vulnerable to spoofing attacks as a registered user's photographs/videos can be gotten with ease through the internet or simply capturing their face using a camera, even without the consent or physical contact with him/her. Hence, the need for the development of anti-spoofing measures against such attacks. Several efforts have been in place in research for the development of face anti-spoofing methods. In this study, authors reviewed various anti-spoofing methods by several authors by searching related papers using keywords from online sources. This paper discusses several approaches to face spoof detection, the different types of face-spoof attack, and provide a taxonomy of the various anti-spoofing methods. The results are presented in a tabular form, providing a comprehensive list of publicly available face anti-spoof databases and a performance comparison table of several approaches.

Keywords—face recognition, anti-spoofing, spoof detection, liveness detection)

I. INTRODUCTION

An ordinary face recognition system void of concrete and deliberate anti-spoofing measures (as the case is with many deployed face recognition systems) is prone to failure if there be spoof attack attempts on such a system. Face recognition is gaining high popularity and application as a biometric authentication system. The face recognition system draws its advantage over other biometric technology such as iris and fingerprints from the fact that it is very convenient, contactless, and non-invasive [1],[2]. It is also one of the biometric information processes with more convenient applicability and a higher working range than others [3]. A report by the International Biometric Group (IBG), shows that facial recognition is next to fingerprint as the most widely deployed biometric data worldwide [4]. Unfortunately, once any biometric data is stolen and replicated, the advantage of such system is overturned instantly. The case is worse with face recognition systems, as a registered user's photographs/videos can be gotten with ease through the internet or simply capturing their face using a camera, even without the consent or physical contact with him/her.

Face spoofing attacks have taken diverse dimensions from printed photos and screen photos up to 3D masks (see Fig. 4). Developing a counter (anti-spoofing) measure against face spoofing is prominent as such, it has raised the interest of researchers in computer vision and biometrics. Efforts have been put in place by several researchers to develop a facial anti-spoofing technique for face recognition systems. This work aims at reviewing various methods proposed for facial anti-spoofing.

Early face anti-spoofing approaches based their cues on the physiological behaviors of the human face such as the blinking of the eyes, head rotation, facial expressions, and lips movement [5]–[7], majorly against print attacks. As spoofing attempts intensifies and diversifies, approaches based on other cues such as Texture analysis, Image Quality Analysis (IQA), and Image Distortion Analysis (IDA) also emerged (see Fig. 6). In most of the efforts reviewed in this paper, authors experimented with their approach on publicly available face anti-spoofing databases (see Table 1) for evaluation and easy comparison. The availability of these face anti-spoofing databases publicly has drastically encouraged many in the development of countermeasures against face spoofing by relieving researchers the stress of data acquisition issues to focus on the design of efficient anti-spoof methods. This paper seeks to:

- Review various methods proposed for face spoof detection in facial recognition systems.
- Identify, define, and group the spoofing mechanisms.
- Provide a comprehensive list of Face Anti-Spoof Databases available in public domains.
- Finally, present a comparative table of the various anti-spoofing techniques reviewed.

II. LITERATURE REVIEW

In literature, several methods for liveness detection have been revealed. Here, several approaches have been reviewed. Earlier approaches were majorly based on motion cues (eye blinking, mouth movement, head rotation, etc.). Such approaches are primarily designed for printed photo attacks [1].

Authors in [2] put forward a method for liveness detection based on eye-blinking using Conditional Random Fields (CRFs). Eye blinking could happen spontaneously. Traditionally, an individual's blinking rate while at rest is 20 times each minute, with typical blinking time that lasts 1/4 second [3]. Hence, the blinking of the eyes can be an awfully effective indication of lives. The experimental results with the samples on cascaded AdaBoost, CRF, and HMM (Hidden Markov Model) reveals CRF did notably better than the rest. Although it attained a liveness detection rate of 98.3%, it is limited by the high response time, computational complexity, and can easily be bypassed by more sophisticated attacks such as fake motion [1].

Singh, Joshi and Sandi in [4] proposed a face recognition system with live face detection based on eye and mouth movement. The authors divided their approach into two parts: live face detection and face recognition. Firstly, the system checks for the liveness of the user, and if the liveness check

is passed, then face recognition is carried out. The movement of the eyes (opened and closed) was done by checking for eye existence within the eye region. If the eye is found, it is assumed that the eye is opened else it is closed. In the same manner, mouth movement is ascertained by checking for the existence of teeth within the mouth region using the HSV (Hue, Saturation, Value) value of the teeth. Detected teeth denote mouth is open else close. Although the system had good accuracy in a few samples of various forms of attacks generated by the authors, the result also shows that the liveness test was bypassed by cut-photo attacks where both the eye and the mouth region are cut out.

Killioğlu, Taşkıran, and Kahraman [5] proposed a method based on pupil tracking. In their approach, they developed a system for observing pupil direction using basic hardware equipment (the square frame that has a total from the eight Light Emitting Diodes, one for each direction). Firstly, the HaarCascade Classifier of the OpenCV library is used to extract eye area from a realtime camera. Kanade-Lucas-Tomasi feature tracker is then used to extract and trace feature points to obtain stable eye regions. The eye region is cropped and rotated to get an eye area that is stable. After which the extraction of the Pupils from the eye area is carried out. When a few numbers of stable frames with pupils are obtained, Arduino is signaled to activate a LED (Light Emitting Diode) from a randomly selected direction. Observations are then made to ascertain whether pupil direction and LED position match. Satisfactory compliance returns data containing liveness information. Although testing of the algorithm on volunteers shows a high success ratio was achieved, the approach has low robustness and generic ability. It can easily be circumvented by a 3D mask or cut (to forge eye blinking and movement) attack. Also, because of the extra hardware requirements, building it into existing face recognition systems will not be easy.

A method proposed by Dhawanpatil and Joglekar[6] seeks to combine Moiré patterns and Image Distortion Analysis (IDA) to detect spoofing attempts. The authors divided the process into 3 modules (see Fig. 1). Input module which accepts the face of an individual which can be a genuine or spoofed face. The spoof detection module, here in identifying whether the face is spoofed or not, the face image is being checked for Moiré patterns and IDA. When you overlay two patterns on each other, Moiré patterns are produced. For a replay attack, light from the device showing the video of the genuine user will cast on the screen of capturing device, thereby producing the Moiré pattern used in detecting the Replay attack. The detection of the Moiré patterns is carried out by the aid of LBP (Local Binary Patterns) descriptor, as a comparison of LBP histograms of captured face with those of the database is carried. Whether the face is spoofed or a real is determined by the result [1]. In the case of printed paper attack, an observation of the surface roughness of the real face and the spoof face is made. Such Moiré patterns are identified in where there are spoof attacks and the hue and saturation parameters of the image are being used for comparison of the color histograms of the genuine and spoofed faces. Image distortion analysis is then carried out and finally, in the face matcher module, the image is being classified as either spoofed or not using an SVM (Support Vector Machine) classifier.

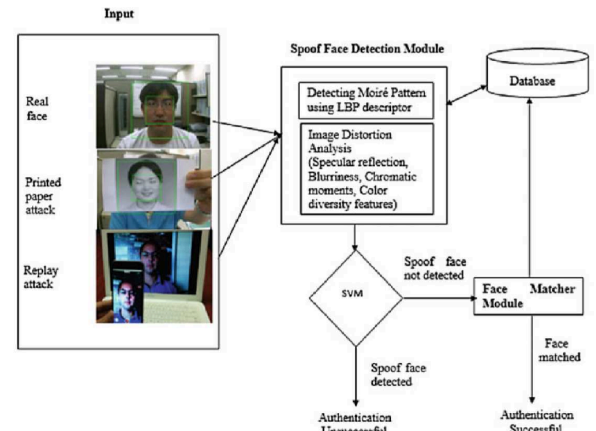


Fig. 1: Block diagram for spoof detection system in [6]

To accomplish the goal of ascertaining how effective depth data can be used to detect spoof attacks, Albakri and Alghowinem[7] developed a system that utilizes the 3D sensor of Microsoft Kinect device for depth information. First-time user's faces and information are first captured. The system scans the image and in parallel measures the depth by placing five points over the scanned image. These five points are marked by placing a dot at the image center and at 30 pixels on the left, right, above, and below the image center point. The depth values are expected to vary and so are being compared. A face is considered 2D and fake where the depth values of at least two point are equal. The detection and recognition of faces achieved by the use of the Haar-Cascade functionalities of the OpenCV library which is compatible with Kinect Sensor. The system was tested with five different spoof test cases. The results show that the system achieved 100% accuracy in detecting spoof cases and thereby denying them access. It was observed that the size of the image and alterations such as wearing glasses, scarf, and makeup did not influence the system, as the system could still detect spoof images and grant access to real users with such alterations. On the other hand, despite that the Kinect sensor's depth range is 0.7 - 6 m, it was observed that test cases of over 1 meter distance reveals drop in accuracy of spoof detection. Also, in a situation that the countermeasure is solely based on depth cues, depth sensors become futile when presented with a 3D mask attack [8].

Galbally and Marcel in[9] presented a new approach that employs general image quality assessment for liveness detection. In their approach, a distorted version of each input image is obtained by applying a low-pass Gaussian kernel filter (variance (σ)=0.5, size=3x3). The system extracts from each image, a total of 14 image quality features. The approach was designed to perform the computation of image quality features before any preprocessing activity. This is in a bid to reduce computational complexity. The classification as either real or fake is done using a Simple Linear Discriminant Analysis (LDA). Two datasets available publicly (Replay-Attack Database and CASIA FAS-Database) were used for the experimental validation. Result obtained was highly competitive compared to other state-of-the-art approaches validated on the same benchmarks. Their findings suggest that general image quality analysis holds highly valuable information for distinguishing real from fake face images; The result shows a decrease in performance in cases of

attacks with higher quality in the case of fixed devices, whereas result from CASIA-FAS DB shows a decrease in error rate (higher performance) with an increase in image resolution (of the acquisition device). The experiment also confirms their assumption that the impact of the Gaussian filter on the real and fake image samples differs.

Another method proposed was based on Image Distortion Analysis (IDA) representing the vector by four extracted features: blurriness, specular reflection, chromatic moment, and color diversity [1]. The authors try to go beyond the efforts of Galbally, Marcel, and Fierrez [10] whose work aimed at building a generic anti-spoofing technique for different biometric modalities, but their approach was trained and tested under intra-database (although cross-validation was used). Authors in [1] improved on the generalization by using two public databases: Replay-Attack DB and CASIA FAS DB, and also extended their effort to collect a new anti-spoofing database called Michigan State University Mobile Face Spoofing Database (MSU MFSD). Ensemble classifier was used by training two SVM (Support Vector Machine) classifiers for each attack type (printed photo attack and the replay attack). Results for intra-database test show that the approach outperformed state-of-the-art approaches. For the cross-database testing, it significantly did better than baseline methods. The fact that only a single frame of image is required reduces the computational complexity.

One of the early works was done in [11] in which Lambertian reflectance was the basis for discrimination of the 2D print attacks from 3D live faces. Authors extracted the latent reflectance features by variational retinex-based technique and the Difference-of-Gaussian (DoG) based approach to be used for classification. Experiment on the NUAA (Nanjing University of Aeronautics and Astronautics) Photograph Imposter Database shows a promising result with Area Under Curve (AUC) of 0.94.

Also, the authors in [12] proposed a method that employs multiple DoG filters. Without prior knowledge of the most discriminatory frequency, in their approach, high-frequency features were extracted by the use of multiple DoG filters from the face images, which forms a redundant feature set. Low-frequency details and noises were removed by setting the Gaussian variance (σ). Representing the inner Gaussian variance with σ_1 and σ_2 for the outer one, they considered four DoG filters: $\sigma_1=0.5$, $\sigma_2=1$; $\sigma_1=1$, $\sigma_2=1.5$; $\sigma_1=1.5$, $\sigma_2=2$; and $\sigma_1=1$, $\sigma_2=2$. The filtered images are then concatenated and fed into SVM to train a final classifier. Experimental results on their CASIA-FASD gave an EER (Equal Error Rate) of 0.17. It was a competitive performance; accuracy was not sufficient for real-life application. Their experiments also show that both imaging quality and spoofing type can have significant influence on the accuracy.

Drawing inspiration from IQA, printing artifacts characterization, and difference in light-reflections, authors [13] were the first to come up with the widely used texture-based approach based on Multi-scale LBP for detection of printed photo attacks. The Multi-scale LBP form the basis for facial image analysis in their approach, and the micro-texture patterns were encoded into an enhanced feature histogram. The features were used as input for the SVM classifier for liveness detection. Experiment on NUAA Photograph

Imposter Database gave a better result with an Area Under Curve (AUC) of 0.99 than the previous leading approach in [11] tested on the same database with an AUC of 0.94. However, this approach may not be robust or generic enough for real-life applications today as it is limited to 2D attacks and can easily be bypassed by 3D attacks.

A recent work by Thippeswamy, Vinutha, and Dhanapal [14] proposed a method based on an ensemble of texture descriptors of local appearance-based methods. In their approach, Local Binary Pattern (LBP), Gradient Directional Pattern (GDP), Gradient Local Ternary Pattern (GLTP), Local Gabor Binary Pattern Histogram Sequence (LGBPHS), Local Directional Pattern (LDiP), and Local Phase Quantization (LPQ) descriptors were concatenated to form an ensemble feature vector representation of images. Classification into real or impostor images is then done using χ -nearest neighbors (χ NN) classifier. Their algorithm was implemented on the NUAA Photo Imposter database. Results show an impressive performance on identifying spoofed faces as 98.39% of spoofed faces were rightly identified. On the other hand, the system's performance in identifying real faces was poor as only 51% of real faces were correctly identified.

In [15], authors examined the RGB, HSV, and YCbCr color spaces in order to understand color space is more discriminative for distinguishing genuine faces from spoofed faces. Color LBP descriptor was used for the analysis of the joint color-texture information from the luminance and the chrominance channels. Their experiments on CASIA FAS DB and Replay-Attack DB, show that the color texture-based approach outperforms that of gray-scale in detection of diverse spoof cases. In comparison of the various color spaces, YCbCr yielded the best performance. The result also shows that HSV color space features tend to be more effective against video attacks than those of YCbCr color space. combining HSV and YCbCr color texture representations gave the overall best performance.

In [8], Boulkenafet, Gölüm, and Hadid extended their preliminary color texture-based approach in [15] and proposed a robust and generalized model based on the observation that the color spaces like HSV and YCbCr offer more discriminating features between real and spoof images than their counterpart RGB channels. In their approach, Local Phase Quantization (LPQ) features in YCbCr and multiscale LBP features of the face were used. Experimental results on three challenging public databases: CASIA-FASD, Replay-Attack DB, and MSU MFSD, revealed an excellent performance. For the CASIA FASD and MSU MFSD, the proposed color texture representation supported the mixture of Color-LBP features in HSV and LPQ features in YCbCr color spaces computed outperformed state-of-the-art, with a competitive result on Replay-Attack DB. Their approach achieved a stable performance across the datasets as opposed to most of the approaches discussed in their literature. The facial color texture representation showed promising generalization capabilities in the cross-database evaluation, hence suggesting that color texture shows more stability in unknown conditions than the grayscale. It is worthy of note that because the optimization of face normalization or the limits of the face bounding box which are shown in [1], [16], [17] to be vital in intra-database tests were not done in their

experiments, a significant effect was also noticed in the cross-database performance.

Considering that existing works based on image distortion analysis, captures the facial details without paying attention to the pattern of the image, Anand and Gupta [18] designed a spoof detection approach that utilizes a hybrid algorithm by combining SIFT(Scale-Invariant Feature Transform) descriptor, genetic algorithm and artificial neural network to create a unique feature sets according to the categories of database images. In their approach, SIFT descriptor was used to extract key points and identify the pattern of the face region. Genetic algorithm was then used as a feature optimization technique for separating unique features and for obtaining a normalized feature set for each category of the image data. Finally, the authors designed an Artificial neural network Classifier for training the feature set. Their experiment was tested on a face spoof database of “JPEG” images constructed by authors using the cameras with non-compression technique. Although the result shows an impressive performance with an average accuracy of 97.86% and an error rate of 2.13%, the system was deficient in rightly identifying real faces as the result shows a False Rejection Rate (FRR) of 53.29%. Fig. 2 below is a diagrammatic representation of their approach.

Another method based on the ensemble of descriptors was proposed in [19]. They combine features extracted from three different feature descriptors, namely: Color moment, Haralick texture, and Color Local Binary Pattern (CLBP) feature descriptors to form the feature space. Features obtained is then classified using Logistic Regression. Performance evaluation on the MSU-MFSD dataset achieved impressive results in comparison to state-of-the-art approaches, with an F1-Score of 99.45%.

In [20], Boulkenafet gomulainen, and Hadid proposed a new approach describes the appearance of the face using Fisher Vector encoding on SURF(Speeded Up Robust Features) features from different color spaces. In their approach, instead of gray-scale representation, SURF features are obtained from color images by applying the SURF descriptor on each color band and then concatenating them form a single feature vector (which they refer to as CSURF). A dimensionality reduction using PCA (Principal Component Analysis) is finally carried out on the face

description after de-correlating the feature vector. The evaluation performances on Replay Attack, CASIA FASD, and MSU datasets was excellent and stable. The cross-database tests outperform state-of-the-art approaches, very promising generic capabilities, despite limited training data used.

A two-stream C $\sigma\sigma$ (Convolution neural networks)-based method was proposed in [21], the method utilizes the full-face image and patches from the same face to distinguish between real and fake faces. C $\sigma\sigma$ uses local features extracted from the image for discriminating fake patches while examining whether the image has a face-like depth is done using the holistic depth map extracted from same image. Experimental result on CASIA-FASD, MSU-USSA, and Replay Attack datasets shows an impressive performance in comparison to previous state-of-the-art performance. Although similar EER to previous approach was obtained, they got much smaller HTER (Half Total Error Rate. Also, the fusion brought about an increase on the AUC from 0.989 to 0.997 for Replay-Attack DB.

Liu, Jourabloo, and Liu [22] proposed approach that coherently combines C $\sigma\sigma$ and R $\sigma\sigma$ (Recurrent neural network) architecture. Unlike other deep learning methods that approached face spoof detection as a binary classification task, their approach, instead of extracting any cues to separate two classes but cannot be generalized, focuses on the known-spoof patterns across spatial and temporal domains. It leverages two auxiliary information as supervision: the depth map and rPPG (Remote Photoplethysmography) signals (see Fig. 3). The C $\sigma\sigma$ part harnesses the depth map supervision to find hidden texture features that ends up in distinct depths for real and fake faces. Rather than the traditional way of computing the rPPG signal [20], the R $\sigma\sigma$ part learns to estimate the rPPG signal, which checks the temporal variability of video frames. An experiment of the model on CASIA-MFSD, Replay-Attack DB, OULU- σ PU, SiW, on both intra-database and cross-database testing attained state-of-the-art results. In comparison with previous leading state-of-the-art results[8], their approach reduces the cross-testing

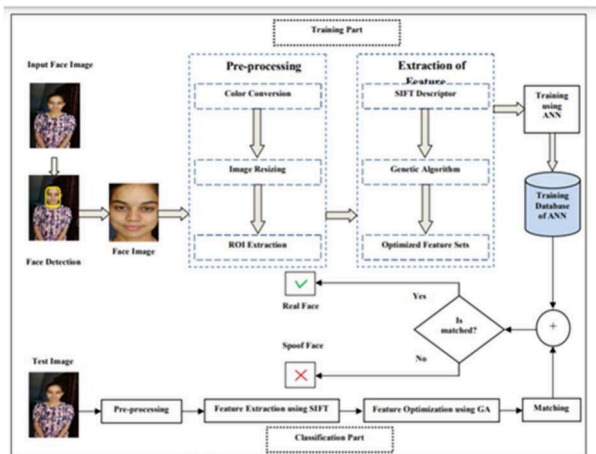


Fig. 2: Structure of face spoof detection system proposed by Anand and Gupta[18]

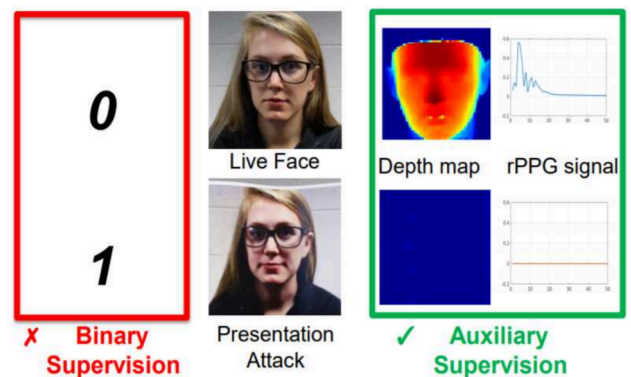


Fig. 3: Binary supervision vs auxiliary supervision[22]

errors (HTER) by 24.6% and 8.9% on the CASIA-MFSD and Replay-Attack datasets respectively.

In previous efforts on depth supervised learning, only in a single frame is depth regarded as auxiliary supervision. In [23], authors proposed a method to estimate depth information from multiple RGB frames. The approach consists of two main modules: The single-frame part, which focuses on seeking cues for spoofing with the static depth supervision; While the second is the multi-frame part, intended to extract both the long-term alongside the short-term motions capable of discriminating real and spoof face: The optical-flow-guided-feature-block(OFFB) serves as the short-term motion module, while convolutional gated recurrent units (ConvGRU) serve to model long-term motion patterns. Experimental result on CASIA-MFSD, OULU-σPU, SiW, and Replay-Attack databases shows that their method attained state-of-the-art results.

In a recent work in [24], authors designed a Multi-Regional Convolutional Neural Networks (MRCNN) as a basis for their method. The approach attempts to harness information from all patches by scattering the gradient distribution, in order to improve how robust, it can detect spoof. Excessive emphasis of certain local areas is avoided by utilizing information from the whole face region. This done by introducing to local patches, the concept of local classification loss. Experiment on Replay Attack, SiW, OULU-σPU and Print Attack datasets shows better performance in terms of robustness against adversarial attacks, and an improvement in performance against traditional attacks in comparison to previous approaches.

Another recent and much interesting work was done in [25]. In an attempt to overcome the weakness of multi-modal approaches in balancing performance and computational complexity, authors proposed a method with streaming module which they refer to as FeatherNet A/B trained by depth data. Authors also offer a novel fusion classifier using "ensemble + cascade" structure. An experiment of their approach on two datasets: CASIA-SURF[26] and their proposed Multi-Modal Face Dataset (MMFD) reveals an excellent performance. Nevertheless, their approach could not overcome the intrusive nature of multi-modal systems.

In a more recent work in [27], authors proposed a multi modal approach based on Central Difference Networks which could capture the intrinsic spoofing patterns for RGB, depth and infrared modalities. Their approach emerged first and second in the "Track Multi-Modal" and the "Track Single-Modal (RGB)" respectively during the ChaLearn Challenge@CVPR2020 [28]

III. FACE SPOOFING ATTACKS

Face recognition systems have security weaknesses which means they can be bypassed or misled using various face spoofing attacks. Spoofing refers to copying or disguising a real person's identity and using the same for harming any type of data [6]. Face spoofing or face presentation attack is an illegal activity where a fraudulent user circumvents a facial recognition system by disguising themselves as a registered user to gain unauthorized access and privileges. Generally, there are two main categories of face spoof attacks: 2D and 3D face attacks as seen in Fig. 4.

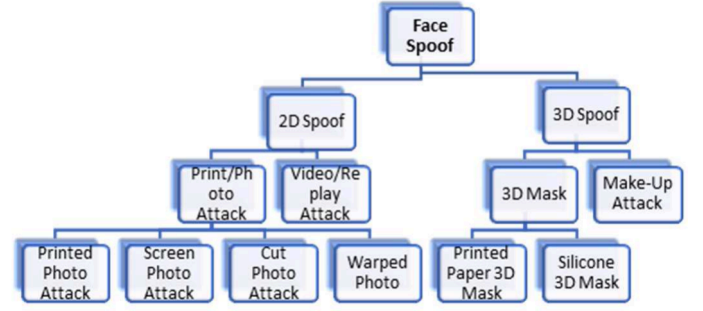


Fig. 4: Different types of face spoof

A. 2D Face Attacks

This consists of face presentation attacks with planar shapes (flat surface). They include:

Print/Photo Attacks: Print presentation attacks are achieved by displaying to face recognition system a photograph of a real user which might have been gotten with a camera or even from the social media images uploaded by the user online [29]. This is one of the most common since most people have photographs on the internet that can be accessed without their permission, thus these photographs could be used to attempt this attack. They are presented in the following ways:

- **Printed-photo Attack:** This involves the utilization of flat printed photos of the genuine user.
- **Screen-photo Attack:** This is achieved by displaying the photo of a genuine user through a digital screen of an electronic device (e.g. mobile phone, tablet, etc.).
- **Cut-photo Attack:** This attack uses a printed copy of real user's photo, where the face regions such as eye, nose, or mouth has been cut out, while another person places his face behind the paper cuts to forge the eye-blinking and movement behavior of the live face.
- **Warped-photo Attack:** Bending a printed photo in different directions to simulate facial motion.

Video attack: This is commonly referred to as a Replay attack. It is more sophisticated than the photo attack in tricking a face recognition system. Here a looped video of a victim's face is displayed using a digital device such as tablets or large smartphones. In this approach, physiological behaviors are more natural. It has Physiological signs of life such as expressions of the face, eye-blink, mouth, and head movements, which are often not there in photos.

B. 3D Attacks

Here, the spoofing artifact is the 3D mask of the face of a real user, this makes it more difficult finding an accurate counter-measures for it. Since the whole 3D structure of the human face is emulated, the employment of depth cues which may be an answer to stop the attacks based on planar surfaces becomes inefficient in the case of 3D attacks[30]. Other forms of 3D attacks include sophisticated makeup and plastic surgery.

IV. TAXONOMY OF FACE ANTI-SPOOFING METHODS

A. Active and Passive Anti-Spoof Methods

Firstly, face anti-spoof techniques can be classified into two broad groups: Active methods and passive methods. The distinction between active and passive anti-spoof methods is

based on the enrollment mode (activation) or mode of capturing the user data.

- a. **Active Anti-Spoof Methods:** This includes methods that involve the active participation of the user in the data capturing or enrolment process for spoof detections. Active face liveness detection is interactive in nature. The user is requested to perform a few actions in front of the camera, for example: to open and close his mouth to demonstrate mouth movement, blink his/her eye, smile, and nodding of the head. The person must complete all actions successfully. Active approaches, have strong protection ability against spoofing but it requires user interaction which makes it intrusive and might not be appropriate for some use cases. For a faster and more convenient anti-spoofing approach, passive liveness detection is employed.
- b. **Passive Anti-spoof Methods:** These are non-intrusive methods of liveness detection. There is no need for user interaction. Usually, users need not know they are being tested. The system performs everything on its own – without the user paying any attention.

B. Classification Based on Liveness Cues

Secondly, classifying face anti-spoof methods based on liveness cues can be as follows:

- Motion
- Texture
- Image Quality
- Depth information
- Multi-modal

- Ensemble of descriptors
- Deep learning approach
- Sensor level approach

V. RESEARCH METHOD

A. Study Selection Procedure

A subject search was carried out on face recognition with spoof detection and its synonyms in six databases: Web of Science, Research Gate, Science Direct, Scopus, IEEE Xplore, and Springer. Using “Face” as the keyword in the search domain helps to preclude any non-face spoof detection like fingerprint and iris. The search scope was also limited to English literature only. The selection included searching the literature source, then screening and filtering obtained articles. Irrelevant and duplicate articles are being removed by checking the title and abstract.

B. Search

Search for literature in the aforementioned databases were done using the keyword “Face Recognition” with the “AND” operator and “Spoof Detection” with different synonyms such as anti-spoof, liveness detection, etc. as seen in Fig. 5 showing the query text.

Query:

(“face”) AND (“anti-spoofing” OR “spoof detection” OR “liveness detection”) refined by

Fig. 5: Search query

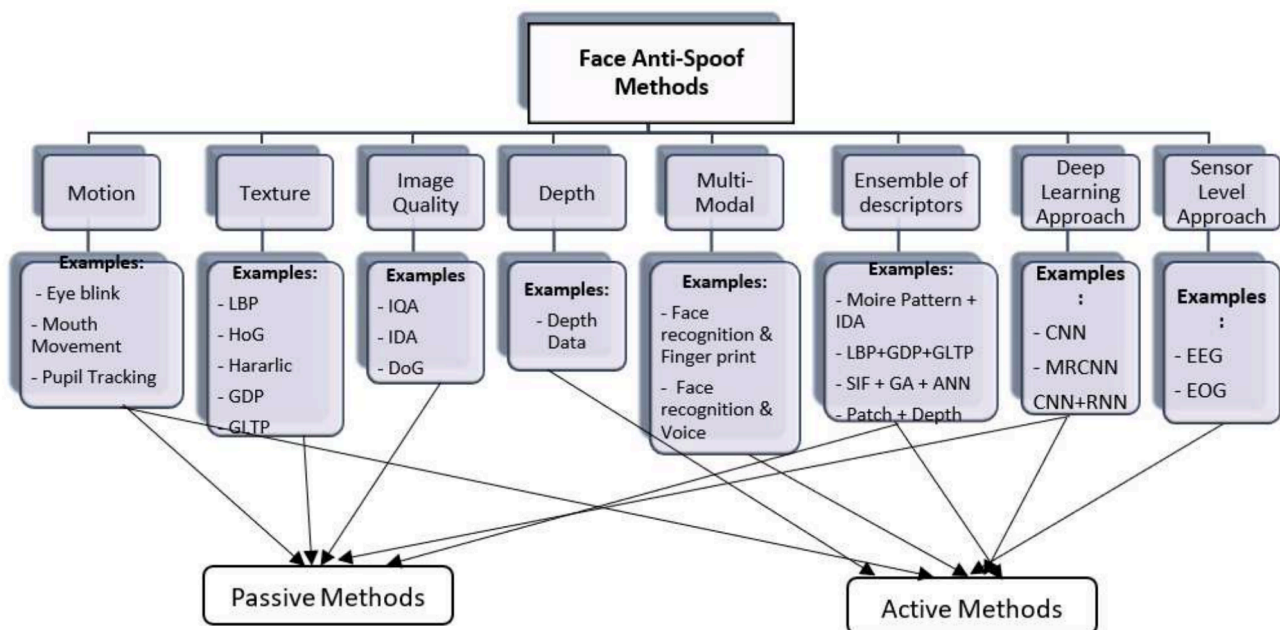


Fig. 6: Diagrammatical summary of face anti-spoofing methods with examples

TABLE I. LIST OF PUBLICLY AVAILABLE FACE ANTI-SPOOF DATASET

S/N	Dataset Name	Year	No. of Subjects	No. of Live/Spoof	File Type	Spoof type
1.	σ UAA [12]	2010	15	5105/7509	Image	Warped photo
2.	PRIoT-ATTACg Database [31]	2011	50	200/200	Video	Print Photos
3.	Yale-Recaptured	2011	10	640/1920	Image	Flat printed phot
4.	IIIT-D Sketch Database [32]	2011	106	4603	Image	Flat sketched images
5.	REPLAY-ATTACg DB [33]	2012	50	200/1000	Video	Print and replay
6.	CASIA-FAS DB [12]	2012	50	150/450	Video	Warped, cut: Replay video
7.	3DMAD [34]	2013	17	30600/45900	Video	Video & 3D mask attacks
8.	gose and Dugelay	2013	20	200/198	Video	3D Mask
9.	MSU MFSD [1]	2015	35	110/330	Video	Print and Replay
10.	UVAD	2015	404	808/16268	Video	Video Replay
11.	BUA Lock3DFace[35]	2015	509	5711	Video	
12.	MSSPOOF [36]	2016	21	4704	Image	Print
13.	REPLAY-MOBILE [37]	2016	40	390/640	Video	Photo and Video Attack
14.	MSU-USSA [38]	2016	1140	1140/9120	Image	print, Replay
15.	SMAD [39]	2016	12	1008	Video	3D silicon mask
16.	OULU- σ PU [40]	2017	55	1980/3960	Video	Print and video/replay
17.	SiW [22]	2018	165	1320/3300	Video	Print and Replay
18.	MFAD [41]	2018	50	1000/1100	Video	Print, Replay
19.	CASIA-SURF CeFA [42]	2019	1,607	23538	Video	Print, Replay, 3D print & Silica mask
20.	CASIA-SURF [26]	2019	1000	3000/18000	Video	Print, Cut
21.	CelebA-Spoof [43]	2020	10177	625,537	Video	Print, Replay, 3D mask
22.	ROSE-Youtu [44]	2020	20	3350	Video	Printed, replay, and 3D mask attacks

TABLE II. PERFORMANCE COMPARISON FOR DIFFERENT PROPOSED METHODS

Study	Method	Dataset	Accuracy
[9]	General IQA	Replay-Attack MSU MFSD	Intra-DB: TPR = 92.2% @FAR= 0.1 for REPLAY ATTACg Cross-DB: TPR = 75.5% @FAR= 0.1 for MSU-MFSD
[18]	SIFT+ GA+ $\Delta\sigma$	Custom dataset	97.86
[19]	Haralick+CLBP +Color Moments	MSU-MFSD	99.45
[22]	C $\sigma\sigma$ +R $\sigma\sigma$ (Dept + rPPG)	SiW, Olulu- σ PU, MSU-MFSD, Replay-Attack	Intra DB: ACER= 3.58 on SiW DB; ACER= 1.6 on OULU- σ PU Cross-DB: HTER= 27.6% for MSU-MFSD & HTER= 28.4% for Replay-Attack
[21]	Patch and Depth-based C $\sigma\sigma$	CASIA-FASD, MSU-USSA, and Replay Attack	AUC = 0.997 on Replay -Attack EER= 2.67%, HTER=2.27% on CASIA-FAS Database EER= 0.79%, HTER= 0.72% on Replay-Attack EER= 0.35 $\pm$ 0.19, HTER= 0.21 $\pm$ 0.21 on MSU-USSA
[12]	DoG	CASIA-FASD	EER = 0.17
[13]	Multi-scale LBPs	σ UAA	AUC = 0.99
[14]	Ensemble of Texture descriptors (LBP+GDP+GLTP+LDiP+LGBPHS+ LPQ)	σ UAA	TRR= 0.98.39 FRR= 0.49
[24]	MRC $\sigma\sigma$	Print Attack DB, Replay Attack DB OULU- σ PU DB and Spoof in the Wild (SiW) DB	HTER = 0.71 on Print Attack Database HTER= 1.6 on Replay Attack DB ACER= 1.9 on OULU- σ PU; ACER= 1.3 on SiW database
[23]	Temporal and depth information	SiW database OULU- σ PU CASIA-MFSD	Intra DB: ACER= 0.73 on SiW; ACER= 1.3 on OULU- σ PU Cross DB: HTER = 17.5 on Repaly Attack DB; HTER = 24.0 on CASIA-MFSD
[17]	C $\sigma\sigma$	Replay Attack CASIA-MFSD	Intra DB: HTER=6.25.68, EER=6.23 on CASIA; HTER=2.68, EER=4.32 on CASIA Cross DB: HTER= 41.36 , EER= 42.48 on Replay Attack HTER= 42.04, EER=41.86 on CASIA MFASD

[20]	CSURF (HSV+YCbCr)	Replay Attack CASIA FASD MSU MFS	EER= 3.3, HTER=8.2 σ Replay Attack EER= 5.7 on CASIA; EER= 7.1 on MSU Inter DB: EER= 35.5
[20]	CSURF (YCbCr)	Replay Attack CASIA FASD MSU MFS	Intra DB: EER= 5.2, HTER=8.9 σ Replay Attack EER= 7.8 on CASIA; EER= 9.2 on MSU Inter DB: EER= 36.7
[20]	CSURF (HSV)	Replay Attack CASIA FASD MSU MFS	Intra DB: EER= 6.2, HTER=11.5 σ Replay Attack EER= 7.1 on CASIA; EER= 7.0 on MSU Inter DB: EER= 43.1
[20]	CSURF (RGB)	Replay Attack CASIA FASD MSU MFS	Intra DB: EER= 11.3, HTER=13.5 σ Replay Attack EER= 14.1 on CASIA; EER= 17.3 on MSU Inter DB: EER= 44.8
[20]	SURF(Gray)	Replay Attack CASIA FASD MSU MFS	Intra DB: EER= 19.5, HTER=21.2 σ Replay Attack EER= 17.8 on CASIA; EER= 18.8 on MSU Inter DB: EER= 45.8
[19]	(Haralick+CLBP +Color Moments)	MSU-MFSD	Precision = 99.45; Recall =99.45; F1-Score =99.45
[45]	EDDTCP	σ UAA MSU-MFSD Replay-Attack	EER = 5.30, HTER=5.74 on σ UAA EER = 2.37, HTER=6.56 σ MSU-MFSD EER = 1.26, HTER=7.86 on Replay Attack
[45]	LBP	σ UAA MSU-MFSD Replay-Attack	EER = 10.54, HTER=12.09 on σ UAA EER = 3.49, HTER=12.54 σ MSU-MFSD EER = 1.96, HTER=10.72 on Replay Attack
[45]	LTP	σ UAA MSU-MFSD Replay-Attack	EER = 9.70, HTER=11.00 on σ UAA EER = 3.50, HTER=12.64 σ MSU-MFSD EER = 0.75, HTER=19.99 on Replay Attack
[25]	Featheronet A/B	CASIA-SURF (MMFD)	ACER=0.0013, TPR=0.999 @FPR=10e-2 TPR=0.998 @FPR=10e-3 TPR=0.9814 @FPR=10e-4
[27]	Central Difference Convolutional networks Multi Modal (Depth+ IR)	CASIA-SURF CeFA	ACER=1.02 $\pm$ 0.59% Track Multi-modal ACER =4.84 $\pm$ 1.79% Track Single-modal

VI. LIMITATIONS AND DISCUSSION

The development of a robust face anti-spoofing method remains a challenging problem. Current approaches still perform below the demands of most real-world applications (especially in terms of generic ability). In the cases of intra-database evaluation, where the disparity between the circumstances of the real faces captured for enrollment and that of the spoofed cases is not much, the performances are impressive. However, most of the hand-crafted features have proven to be limited in terms of generalization ability. It is seen in TABLE II that all the performances in the intra-database scenarios are far better than that of the cross-database.

To tackle all spoof cases, two current trends with promising results—deep learning-based approaches and the combination of multiple cues. However, another limitation of concern is the availability of datasets. Due to the difficulties of creating datasets and the continuous evolution in deploying increasingly sophisticated attacks, anti-spoof methods might be faced with spoof cases not incorporated in the training dataset—a situation called “Unknown Attacks” is challenging. Face anti-spoof datasets are still limited, both in terms of volume and varying types of spoof cases and the devices used to acquire them.

There is still no public large-scale face anti-spoof dataset in the wild. While there exist many of such dataset for face recognition. Of course, the lack of voluminous and diverse

datasets also contributes to the limited enforcement of the current face anti-spoof approaches. Especially the recent deep learning-based approaches, which depends on a large volume of data—is affected by the volume and diversity of dataset for training. Hence, learning features capable of discriminating between a real-face and all kinds of spoof case, and possibly under varying capture conditions, is still open to research.

VII. CONCLUSION

This review work discussed different face anti-spoofing methods proposed so far. Different approaches were considered, with their features. The various forms of face spoof attacks were also discussed. The various anti-spoofing methods were also classified based on enrollment mode and based on liveness detection cues. This study also presented a comprehensive list of publicly available face spoof databases and a performance comparison table of the various face anti-spoof methods.

Although the struggle for a technique with higher accuracy and more generalized ability continues, recent approaches have attained a considerable high accuracy and generalized capability. The study also shows prospects in using color texture, especially the YCbCr color channels and deep learning approaches to obtain the desired output.

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